

Topic : KINEMATICS
Time: 1 Hr Marking +4 -1

- A stone thrown down with a speed u takes a time t_1 to reach the ground, while another stone, thrown upwards from the same point with the same speed, takes time t_2 . The maximum height the second stone reaches from the ground is :

(A) $\frac{1}{2}gt_1t_2$ (B) $\frac{1}{8}g(t_1 + t_2)^2$ (C) $\frac{1}{8}g(t_1 - t_2)^2$ (D) $\frac{1}{2}gt_2^2$
- Two stones of mass m_1 and m_2 (such that $m_1 > m_2$) are dropped Δt time apart from the same height towards the ground. At a later time t the difference in their speed is ΔV and their mutual separation is ΔS . While both stones are in flight:

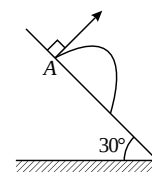
(A) ΔV decreases with time and ΔS increases with time
 (B) Both ΔV and ΔS increase with time
 (C) ΔV remains constant with time and ΔS decreases with time
 (D) ΔV remains constant with time and ΔS increases with time
- A narrow but tall cabin is falling freely near the earth's surface. Inside the cabin, two small stones A and B are released from rest (relative to the cabin). Initially A is much above the centre of mass and B much below the centre of mass of the cabin. A close observation of the motion of A and B will reveal that:

(A) both A and B continue to be exactly at rest relative to the cabin
 (B) A moves slowly upward and B moves slowly downward relative to the cabin.
 (C) both A and B fall to the bottom of the cabin with constant acceleration due to gravity
 (D) A and B move slightly towards each other vertically
- The co-ordinates of a moving particle at any time ' t ' are given by $x = \alpha t^3$ and $y = \beta t^3$. The speed of the particle at time ' t ' is given by :

(A) $3t\sqrt{\alpha^2 + \beta^2}$ (B) $3t^2\sqrt{\alpha^2 + \beta^2}$ (C) $t^2\sqrt{\alpha^2 + \beta^2}$ (D) $\sqrt{\alpha^2 + \beta^2}$
- Two bodies are projected from the same point with equal speeds in such direction that both start the same point on a plane, whose angle of inclination is ' α '. If first body is projected at angle ' θ ' with the horizontal, then the ratio of their time of flight is :

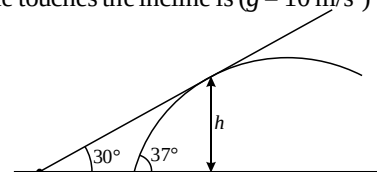
(A) $\frac{\cos(\theta - \alpha)}{\cos \beta}$ (B) $\frac{\sin(\theta - \alpha)}{\cos \theta}$ (C) $\frac{\sin(\theta - \alpha)}{\sin \beta}$ (D) $\frac{\cos(\theta - \alpha)}{\sin \theta}$
- A ball is projected from point A with a velocity 10 m/s perpendicular to the plane as given. The range of the ball on plane is :

(A) $\frac{40}{3}$ m (B) $\frac{20}{13}$ m
 (C) $\frac{13}{20}$ m (D) $\frac{13}{40}$ m



7. A particle is projected with speed 50 ms^{-1} at an angle 37° with the horizontal and it touches an inclined plane of inclination 30° as shown in figure. Height of the point from the ground where particle touches the incline is ($g = 10 \text{ m/s}^2$) :

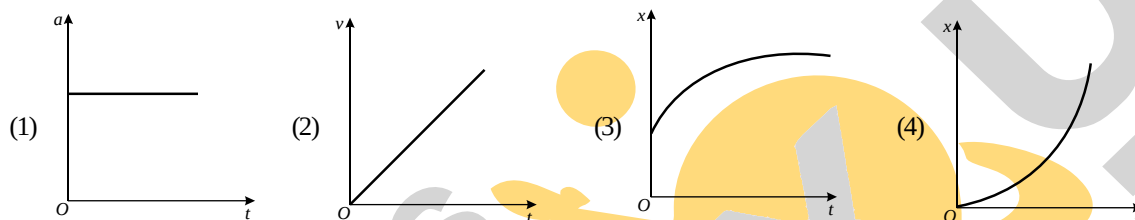
- (A) $\frac{55}{3} \text{ m}$ (B) 55 m
(C) 40 m (D) 18 m



8. A particle is moving in x - y plane with $\vec{v} = v_x \hat{i} + v_y \hat{j}$ and acceleration $\vec{a} = a_x \hat{i} + a_y \hat{j}$. If the magnitude of velocity of the particle is constant then choose the correct option :

- (A) $a_x v_x + a_y v_y = 0$ (B) $a_x v_x - a_y v_y = 0$ (C) $\frac{a_x}{v_x} + \frac{a_y}{v_y} = 0$ (D) $\frac{a_x}{v_x} - \frac{a_y}{v_y} = 0$

9. A particle starts from origin O from rest and moves with a uniform acceleration along the positive x -axis. Identify all figures that correctly represent the motion qualitatively. (a = acceleration, v = velocity, x = displacement, t = time) :



- (A) (1), (2), (3) (B) (1) (C) (1), (2), (4) (D) (2), (3)

10. The position of a particle as a function of time t , is given by $x(t) = at + bt^2 - ct^3$ where a , b and c are constants. When the particle attains zero acceleration, then its velocity will be :

- (A) $a + \frac{b^2}{4c}$ (B) $a + \frac{b^2}{c}$ (C) $a + \frac{b^2}{2c}$ (D) $a + \frac{b^2}{3c}$

11. A particle is moving with speed $v = b\sqrt{x}$ along positive x -axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$) :

- (A) $\frac{b^2 \tau}{4}$ (B) $\frac{b^2 \tau}{2}$ (C) $b^2 \tau$ (D) $\frac{b^2 \tau}{\sqrt{2}}$

12. A car goes from A to B (without changing direction), during first part of journey its average speed is V_1 and for second part of journey average speed is V_2 . The ratio of path length of second part to the path length of first part is $\sqrt{\frac{V_2}{V_1}}$. In

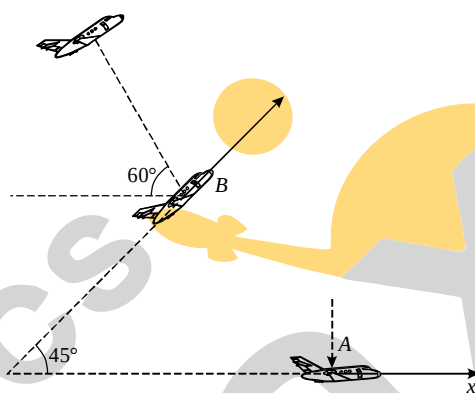
that case the average speed of total path is of average speeds of both parts. Choose correct option of fill.

- (A) Arithmetic mean (B) Geometric mean
(C) Harmonic mean (D) Information is insufficient to answer

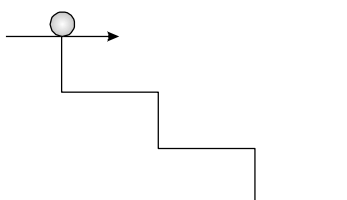
13. Two balls are thrown simultaneously from a top of a building with same initial velocity u . One ball is thrown vertically upwards and the other ball is thrown vertically downwards. The distance between the two bodies after time 5 s is (take, $g = 10 \text{ m/s}^2$) :

- (A) $10u$ (B) $5u$ (C) $15u$ (D) $20u$

14. A body is falling freely from a point A at a certain height from the ground and passes through points B , C and D (vertically as shown) so that $BC = CD$. The time taken by the particle to move from B to C is 2s and from C to D is 1s. Time taken to move from A to B in seconds is :
 (A) 0.6 (B) 0.5 (C) 0.2 (D) 0.4
15. A car accelerates from rest in the 1st part of its journey with $a \text{ m/s}^2$ and then it retards with $2a \text{ m/s}^2$ so as to bring it to rest again. If t is the total time of journey, then the distance travelled by car is :
 (A) $\frac{at^2}{3}$ (B) $\frac{at^2}{2}$ (C) at^2 (D) $\frac{at^2}{4}$
16. Passengers in the jet transport A flying east at a speed of 800 kmh^{-1} observe a second jet plane B that passes under the transport in horizontal flight. Although the nose of B is pointed in the 45° north east direction, plane B appears to the passengers in A to be moving away from the transport at the 60° angle as shown. The true velocity of B is :



- (A) 586 kmh^{-1} (B) $400\sqrt{2} \text{ kmh}^{-1}$ (C) 717 kmh^{-1} (D) 400 kmh^{-1}
17. Two balls A and B are projected simultaneously from the origin with initial velocities 50 m/s and 20 m/s at angle 60° and 30° respectively, with horizontal. The position of the ball A with respect to ball B after 3s in y -direction is given by :
 (A) 117.79m (B) 112.30m (C) 94.50m (D) 100m
18. A person climbs up a stalled escalator in 60s. If standing on the same but escalator running with constant velocity he takes 40s. How much time is taken by the person to walk up the moving escalator?
 (A) 37 s (B) 27 s (C) 24 s (D) 45 s
19. A staircase contains three steps each 10 cm high and 20 cm wide as shown. The minimum horizontal velocity of a ball rolling off the uppermost plane so as to hit directly the lowest plane is:



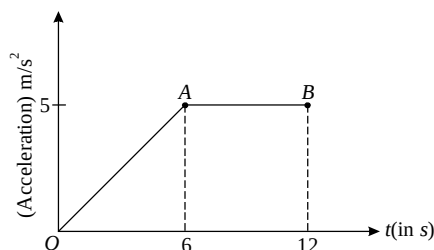
- (A) 1 m/s (B) 2 m/s (C) 3 m/s (D) 4 m/s

20. A projectile is thrown with an initial velocity of $(a\hat{i} + b\hat{j})$ m/s. If the range of projectile is twice the maximum height reached by it, then :

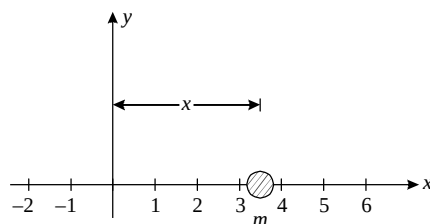
(A) $b = \frac{a}{2}$ (B) $b = a$ (C) $b = 2a$ (D) $b = 4a$

INTEGERTYPE

21. A ball is projected with initial velocity u making angle θ with horizontal and its range is found to be R . Now if the ball is dropped and its velocity after travelling distance R is u , find θ in degrees.
22. A boat crossing a river moves with a velocity v relative to still water. The river is flowing with a velocity $\frac{v}{2}$ with respect to the bank. The angle with respect to the flow direction with which the boat should move to minimize the drift is: (In degree)
23. A swimmer can swim in still water with a speed of $\sqrt{5}$ m/s. While crossing a river his average speed is 3 m/s. If he crosses the river in the shortest possible time, what is the speed of flow of water in SI unit?
24. A particle is projected with initial velocity $u = 10$ m/s from a point $(10\text{ m}, 0\text{ m}, 0\text{ m})$ along z -axis. At the same time, another particle was also projected with velocity $v(\hat{i} - \hat{j} + \hat{k})$ m/s from $(0\text{ m}, 10\text{ m}, 0\text{ m})$. It is given that they collide at some point in the space. Then find the value of v . [$g = 10\text{ m/s}^2$ ($-\hat{k}$)]
25. The velocity of projection of a body is increased by 2%, other factors remaining unchanged, what will be the percentage change in the maximum height attained.
26. A disc of radius 50 cm is projected from the ground with its plane vertical, such that the velocities of the topmost point (A) and lowest point (B) are :
 $\vec{v}_A = 20\hat{i} + 10\hat{j}$ m/s, $\vec{v}_B = 10\hat{i} + 10\hat{j}$ m/s
 at the instant of projection. Here, $\hat{i}$ and $\hat{j}$ are unit vectors in the plane of the disc, along the horizontal and vertically upward directions respectively. The range of the disc is (in metre):
27. If a body loses half of its velocity on penetrating 3 cm in a wooden block, then how much will it penetrate more before coming to rest (in cm)?
28. For an airplane to take-off it accelerates according to the graph shown and takes 12 s to take-off from the rest position. The distance travelled by the airplane is : (in metre)



29. Position of a particle moving rectilinearly over x -axis is given by $x = t^3 - 6t^2 + 9t + 2$ m. How much distance is covered by particle in first 6s? (answer in metre)



30. A ball rolls up a slope. At the end of three second its velocity is 20 cm/s, at the end of eight second its velocity is 0. What is the average acceleration from the third to the eighth second (answer in cm/s^2)?

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Topic : KINEMATICS**ANSWER KEY**

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|---------|---------|---------|---------|
| 1. (B) | 2. (D) | 3. (B) | 4. (B) |
| 5. (B) | 6. (A) | 7. (A) | 8. (A) |
| 9. (C) | 10. (D) | 11. (B) | 12. (B) |
| 13. (A) | 14. (B) | 15. (A) | 16. (C) |
| 17. (D) | 18. (C) | 19. (B) | 20. (C) |

INTEGRITYTYPE

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|-----------|----------------------------|-------------|------------|
| 21. [15°] | 22. [120°] | 23. [2 m/s] | 24. [10] |
| 25. [4%] | 26. [30m] | 27. [1 cm] | 28. [210m] |
| 29. [62m] | 30. [4 cm/s ²] | | |